Chapter 8

Implicit Curves and Conics

A function assigns one output to each input, and its graph is a curve that passes the vertical line test. Most curves in the plane are not of this kind. A circle, an ellipse, the orbit of a planet, the locus where two quantities balance, are described not by solving for y but by an equation that x and y satisfy together. Such a curve is *implicit*: the equation states a condition, and the curve is the set of all points meeting it.

8.1 Curves Defined by an Equation

An equation in x and y singles out the points whose coordinates satisfy it. That set of points is a curve in the plane, whether or not y can be isolated.

Definition: Implicit curve. An *implicit curve* is the set of all points (x, y) satisfying an equation $F(x, y) = 0$. The equation need not determine y as a function of x; a vertical line may meet the curve in several points.

The unit circle $x^2 + y^2 = 1$ is the standard example. No single function describes it, since most vertical lines through the interior cross it twice. Solving for y yields $y = \pm\sqrt{1 - x^2}$, two functions, the upper and lower semicircles, that together reconstruct the curve. The implicit equation holds both halves at once.

Key idea. An implicit equation describes a curve as a condition its points satisfy, not as a rule producing y from x. When y cannot be isolated as a single function, the curve simply is not the graph of a function, and the implicit form is the honest description.

Example 8.1.1. *(testing membership)* Does $(3, -4)$ lie on $x^2 + y^2 = 25$? Substitute: $9 + 16 = 25$, true, so the point is on the curve. Does $(2, 3)$? $4 + 9 = 13 \neq 25$, so it is not. Membership is checked by substitution into the defining equation, never by solving anything.

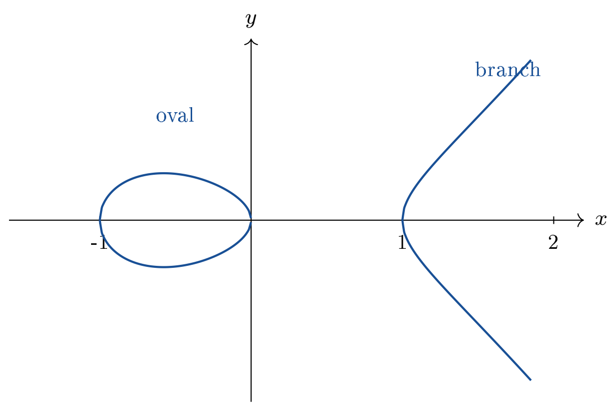
8.2 An Elliptic Curve

A richer example shows how much structure an implicit equation can carry. The equation

$$y^2 = x^3 - x$$

defines an *elliptic curve*. Reading it as a condition, a point (x, y) lies on the curve when its y-coordinate squared equals $x^3 - x$. Since a square is never negative, real points occur only where $x^3 - x \geq 0$, that is where $x(x-1)(x+1) \geq 0$: on the interval $-1 \leq x \leq 0$ and on $x \geq 1$.

The picture. The real locus of $y^2 = x^3 - x$ falls into two separate pieces: a bounded oval over $-1 \leq x \leq 0$, and an unbounded branch opening to the right over $x \geq 1$. Between 0 and 1 the right-hand side is negative, so no real point exists, and the curve has a genuine gap. Such a disconnected shape can never be the graph of a function.



Example 8.2.1. *(points on the elliptic curve)* The points with $x = -1$, $x = 0$, and $x = 1$ all give $y^2 = 0$, so $(-1, 0)$, $(0, 0)$, and $(1, 0)$ lie on the curve; these are where it meets the x-axis. At $x = 2$, $y^2 = 8 - 2 = 6$, so $(2, \sqrt{6})$ and $(2, -\sqrt{6})$ are both on the curve, a vertical line meeting it twice. At $x = \frac{1}{2}$, $x^3 - x = \frac{1}{8} - \frac{1}{2} = -\frac{3}{8} < 0$, so no real point sits above $x = \frac{1}{2}$, confirming the gap.

Common pitfall. The symmetry $y \mapsto -y$ in $y^2 = x^3 - x$ means every point has a mirror image across the x-axis. This is exactly why the curve is not a function: each admissible x with $x^3 - x > 0$ produces two y-values, one positive and one negative.

8.3 Conic Sections

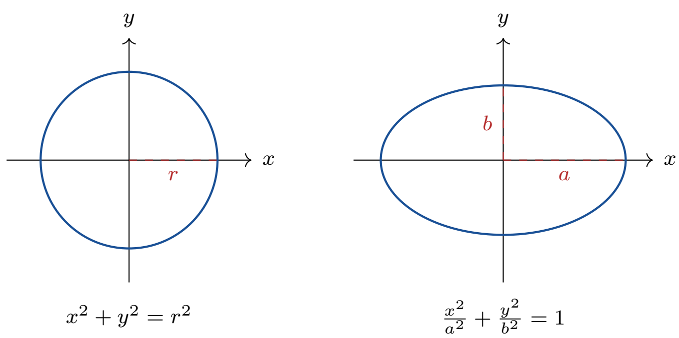
The most important implicit curves are the *conics*, the curves cut from a cone by a plane and, equivalently, the curves defined by second-degree equations in x and y. Four shapes arise: the circle, the ellipse, the parabola, and the hyperbola. Each has a standard implicit equation in which the parameters read off directly as geometric measurements.

Key idea. **The conic sections** are the curves of second degree. In standard position, centered or with vertex at a convenient point:

- *Circle:* $x^2 + y^2 = r^2$, radius r.
- *Ellipse:* $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$, semi-axes a, b.
- *Parabola:* $y = \dfrac{1}{4p}x^2$, or $x^2 = 4py$, focus distance p.
- *Hyperbola:* $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$, opening left and right.

8.3.1 Circle and ellipse

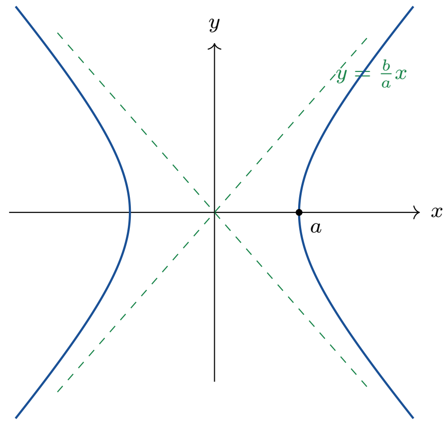
The circle is the locus of points at fixed distance r from a center; placing the center at the origin gives $x^2 + y^2 = r^2$ from the distance formula. Stretching the circle by different factors along the two axes produces the ellipse: dividing x by a and y by b before squaring turns the radius-one circle into $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$, whose graph reaches $\pm a$ along the x-axis and $\pm b$ along the y-axis.



Example 8.3.1. *(reading an ellipse)* The equation $\dfrac{x^2}{9} + \dfrac{y^2}{4} = 1$ has $a^2 = 9$ and $b^2 = 4$, so $a = 3$ and $b = 2$. The ellipse stretches to ± 3 along the x-axis and ± 2 along the y-axis. Setting $y = 0$ gives $x = \pm 3$; setting $x = 0$ gives $y = \pm 2$, confirming the intercepts.

8.3.2 Hyperbola

Changing the plus sign in the ellipse to a minus produces the hyperbola $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$. The curve has two branches opening left and right, and it approaches the diagonal asymptotes $y = \pm\dfrac{b}{a}x$ at the ends, because for large x the constant 1 becomes negligible and the equation approaches $\dfrac{x^2}{a^2} = \dfrac{y^2}{b^2}$.



Example 8.3.2. *(asymptotes of a hyperbola)* For $\dfrac{x^2}{16} - \dfrac{y^2}{9} = 1$, $a^2 = 16$ and $b^2 = 9$, so $a = 4$, $b = 3$. The vertices sit at $(\pm 4, 0)$ and the asymptotes are $y = \pm\frac{3}{4}x$. The branches hug those lines as x grows.

8.3.3 Recognizing a conic from its equation

A second-degree equation with no xy term is a conic, and the signs of the squared terms name which one. Equal positive coefficients on x^2 and y^2 give a circle; unequal positive coefficients give an ellipse; opposite signs give a hyperbola; a single squared term gives a parabola.

Key idea. Identifying a conic from $Ax^2 + Cy^2 + \cdots = 0$ (no xy term):

- $A = C$ (same sign): circle.
- $A \neq C$, same sign: ellipse.
- A, C opposite signs: hyperbola.
- exactly one of A, C is zero: parabola.

Example 8.3.3. *(classifying by inspection)* $4x^2 + 4y^2 = 36$ has equal coefficients, a circle (radius 3 after dividing). $x^2 + 9y^2 = 9$ has unequal positive coefficients, an ellipse. $x^2 - y^2 = 1$ has opposite signs, a hyperbola. $y - x^2 = 0$ has a single squared term, a parabola.

8.4 Exercises

Exercises 8.4

8.4.1. Decide by substitution whether each point lies on $x^2 + y^2 = 25$: (a) $(0, 5)$; (b) $(3, 4)$; (c) $(-4, 3)$; (d) $(2, 2)$.

8.4.2. Solve $x^2 + y^2 = 9$ for y, obtaining two functions, and state the domain of each. Explain how they recombine to the full circle.

8.4.3. For the elliptic curve $y^2 = x^3 - x$, find all points with $x = -1, 0, 1$, and determine whether any real point has $x = \frac{1}{4}$.

8.4.4. Explain, using the symmetry $y \mapsto -y$, why $y^2 = x^3 - x$ is not the graph of a function, and describe its two pieces.

8.4.5. Find every point on $y^2 = x^3 - x$ with $x = 3$, and verify each by substitution.

8.4.6. Write the equation of the circle centered at the origin passing through $(6, 8)$, and state its radius.

8.4.7. Identify a and b for $\dfrac{x^2}{25} + \dfrac{y^2}{16} = 1$, give the four axis intercepts, and sketch the ellipse.

8.4.8. For $\dfrac{x^2}{9} - \dfrac{y^2}{4} = 1$, give the vertices and the equations of the asymptotes, and sketch both branches with the asymptotes.

8.4.9. Classify each conic from its equation: (a) $3x^2 + 3y^2 = 12$; (b) $4x^2 + y^2 = 16$; (c) $y^2 - x^2 = 1$; (d) $x - 2y^2 = 0$.

8.4.10. Explain why the hyperbola $\dfrac{x^2}{a^2} - \dfrac{y^2}{b^2} = 1$ approaches the lines $y = \pm\dfrac{b}{a}x$ for large x, reasoning from the equation.

8.4.11. Starting from the unit circle $x^2 + y^2 = 1$, explain how dividing x by a and y by b before squaring produces an ellipse, and what a and b measure on the resulting curve.

8.4.12. A second-degree equation $Ax^2 + Cy^2 = 1$ has $A = 4$, $C = -1$. Name the conic, give its vertices, and state its asymptotes.